

# Radioactive Decay

NE 630

## Lesson 03

**Prepare:** Read FNRP Sections 1.6–1.7 and define the key terms before class.

### Macro objective

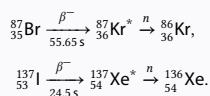
Determine the amounts of a radioactive species and its daughters at any point in time from initial inventories, production rates, and decay constants.

**Vocabulary:** radioactivity;  $\lambda$ ; half-life; mean lifetime; activity; Bq; Ci; initial-value problem; saturation; steady state; secular equilibrium

### 1 Why radioactive species appear

#### Fission-product bridge from Lesson 2

Fission fragments are neutron rich. Their  $\beta^-$  decays produce delayed  $\beta$ ,  $\bar{\nu}$ , and  $\gamma$  energy. When a daughter state is formed above its neutron-separation energy, one decay branch can emit a delayed neutron:



Delayed neutron energies are typically .....  
Deposited  $\beta$  and  $\gamma$  energy drives spent-fuel decay heat; antineutrino energy escapes.

### 2 The single-nuclide decay law

#### From a short-time probability to $N(t)$

For  $N(t)$  radioactive nuclei,

$$\Delta N \approx -\lambda N(t) \Delta t \implies \boxed{N' = -\lambda N}, \quad N(0) = N_0.$$

Separate and integrate:

$$\frac{dN}{N} = -\lambda dt, \quad \ln \frac{N(t)}{N_0} = \dots\dots\dots,$$

$$\boxed{N(t) = \dots\dots\dots}$$

Here  $\lambda$  is the decay probability per unit time for one nucleus.

#### Half-life check

$$\frac{N(nt_{1/2})}{N_0} = \dots\dots\dots, \quad \frac{A(nt_{1/2})}{A_0} = \dots\dots\dots$$

What assumptions make both ratios equal to  $2^{-n}$ ?

### 3 Clocks, number, mass, and activity

#### Fixed relations and unit conversions

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{1}{\bar{t}}, \quad \boxed{A(t) = \lambda N(t)}.$$

**number-mass**  $N = (m/M)N_A$   
**activity** decays per unit time  
**Bq**  $1 \text{ Bq} = 1 \text{ s}^{-1}$   
**Ci**  $1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq}$

When there is no production,  $A(t) = -dN/dt$ . Thus, for a pure radionuclide,

$$m \xrightarrow{\div M} \text{mol} \xrightarrow{\times N_A} N \xrightarrow{\times \lambda} A.$$

#### Reverse the conversion route

Express the radionuclide mass in terms of measured activity:

$$m = \dots\dots\dots$$

List the units you would use for each quantity.

### 4 Production and saturation

#### Decay with a source and saturation

If nuclei are added at rate  $R(t)$ ,

$$\boxed{N' = -\lambda N + R(t)}, \quad N(t_0) \text{ given.}$$

The integrating factor  $e^{\lambda t}$  gives

$$\frac{d}{dt}(Ne^{\lambda t}) = R(t)e^{\lambda t},$$

$$\boxed{N(t) = N(t_0)e^{-\lambda(t-t_0)} + \int_{t_0}^t R(t')e^{-\lambda(t-t')} dt'}.$$

For  $R(t) = R_0$  and  $t_0 = 0$ ,

$$\boxed{N(t) = N_0 e^{-\lambda t} + \frac{R_0}{\lambda} (1 - e^{-\lambda t})}.$$

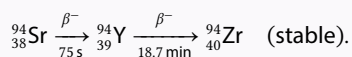
For  $N_0 = 0$ ,

$$A(t) = \dots\dots\dots, \quad N_{\text{sat}} = \dots\dots\dots, \quad A_{\text{sat}} = \dots\dots\dots$$

After  $n$  half-lives,  $A/A_{\text{sat}} = 1 - 2^{-n}$ ; one, three, five, and seven half-lives give 50%, 87.5%, 96.9%, and 99.2% of saturation.

## 5 The $A = 94$ fission-product chain

### Chain and initial data from the notes



Assume  $N_{\text{Sr}}(0) = N_0$ ,  $N_{\text{Y}}(0) = N_{\text{Zr}}(0) = 0$ , and no production after  $t = 0$ .

### Balance each nuclide

$$N'_{\text{Sr}} = -\lambda_{\text{Sr}} N_{\text{Sr}},$$

$$N'_{\text{Y}} = \lambda_{\text{Sr}} N_{\text{Sr}} - \lambda_{\text{Y}} N_{\text{Y}},$$

$$N'_{\text{Zr}} = \lambda_{\text{Y}} N_{\text{Y}}.$$

Label each term as a gain or loss. Why can this one-way chain be solved from the parent downstream?

### Sequential solution

$$\begin{aligned} N_{\text{Sr}}(t) &= N_0 e^{-\lambda_{\text{Sr}} t}, \\ N_{\text{Y}}(t) &= \frac{\lambda_{\text{Sr}} N_0}{\lambda_{\text{Y}} - \lambda_{\text{Sr}}} (e^{-\lambda_{\text{Sr}} t} - e^{-\lambda_{\text{Y}} t}), \\ N_{\text{Zr}}(t) &= \int_0^t \lambda_{\text{Y}} N_{\text{Y}}(t') dt' \\ &= N_0 - N_{\text{Sr}}(t) - N_{\text{Y}}(t). \end{aligned}$$

The last line also follows from conservation of the  $A = 94$  nuclei.

### Inventory after one hour

At  $t = 1$  h,

$$\frac{N_{\text{Sr}}}{N_0} = \text{_____}, \quad \frac{N_{\text{Y}}}{N_0} = \text{_____}, \quad \frac{N_{\text{Zr}}}{N_0} = \text{_____}.$$

Which half-life controls the long-time approach to stable zirconium?

## 6 General chains and equilibrium

### One-way chain with production

For species  $i = 0, 1, \dots$ ,

$$N'_i = -\lambda_i N_i + \lambda_{i-1} N_{i-1} + R_i(t), \quad \lambda_{-1} N_{-1} \equiv 0.$$

Read each equation as

rate of change = parent gain + external gain – decay loss.

Initial values  $N_i(0)$  complete the system. Solve the first equation, insert it into the second, and continue downstream. Branching ratios multiply the corresponding parent-gain terms.

### FNRP extension: secular equilibrium

For a parent 1 and daughter 2 with  $N_1(0) = N_{1,0}$ ,  $N_2(0) = 0$ , no direct sources, and a 100% parent-to-daughter branch,

$$N_2(t) = \frac{\lambda_1 N_{1,0}}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}).$$

If  $\lambda_1 \ll \lambda_2$ , then after several daughter half-lives but before the parent changes appreciably,

$$A_2(t) \approx \text{_____}, \quad \frac{N_2}{N_1} \approx \text{_____}.$$

Explain why equal activities do not imply equal inventories.

### Steady state versus secular equilibrium

**Steady state:** production balances removal for one species, so  $N' = 0$  and its inventory is constant.

**Secular equilibrium:** a short-lived daughter tracks a much longer-lived parent, so their activities are approximately equal while both may vary slowly.

### Choose the model before solving

|                        |                              |
|------------------------|------------------------------|
| no source, one species | $N' = -\lambda N$            |
| source, one species    | $N' = -\lambda N + R(t)$     |
| one-way chain          | coupled gain–loss equations  |
| long-lived parent      | test for secular equilibrium |